

Mark Rothlisberger
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Summary

Staff+ software engineer and technical leader with more than 15 years of experience spanning a wide range of industries and domains, from classified government systems, to incubating start-up concepts within established companies, to building and operating critical large scale payments systems. Trusted cross-functional collaborator and dedicated mentor.

Professional Experience

Freelance Software Consultant

Stealth Startup

August 2026-Present

- One of two engineers building an MVP for a business-to-business marketplace, focusing on financial flows. Using leading-edge agentic workflows to deliver features in days instead of weeks.

Stripe

Staff Software Engineer, Stripe Capital

March 2026-August 2026

- Engineering lead for a project that harmonized and standardized multiple partner integrations to unlock scale through international growth, leveraging AI-native planning tools and development practices to deliver value immediately upon starting role, with minimal ramp-up time.
- Regularly facilitated group-wide operational review to identify common incident themes and increase operational excellence. Pioneered a quarterly high-level review of incidents, using AI to identify cross-cutting themes.

Block (Square)

Group Tech Lead, Payment Methods

2024-2026

- Primary duties included increasing the engineering velocity of 25 engineers across three sprint teams by proactively finding ways to improve engineering designs, drive principled AI adoption, encourage collaboration, and mentor senior engineers.
- Payments eng lead for a zero-to-one international expansion concept; collaborated with product, business, and design disciplines to align on strategy, define scope, evaluate vendors, build prototypes and production systems within tight resource and time constraints.
- Co-authored a strategy document outlining best practices for incorporating LLM-assisted coding output into production systems, with a goal of increasing velocity while maintaining reliability.
- Unblocked a long-stalled, high priority project by identifying a workable path forward and collaborating across multiple engineering disciplines and teams to implement the required features.
- Acted as interim Engineering Manager for a sprint team for four months. Mentored and directed a team of eight engineers, drove increased focus on multiple major projects involving core payment systems, and shepherded high-priority unplanned work to successful completion.
- Directly Responsible Individual (DRI) for Payment Availability SLO, tracking and reporting on performance of critical, complex, high-volume, high-availability systems. These techniques influenced the definition of the first company-wide contractual SLAs for enterprise customers.

Senior Staff Software Engineer, Alternative Payment Methods

2022-2024

- Engineering lead from conception to release for Square Dev AI, one of the first Generative AI tools produced by Block. The tool answers questions posed by external developers in multiple forums,

using a LangChain RAG pipeline populated with developer documentation. System design prioritized correctness over speed. Led a cross-functional team of six people, including machine learning engineers, software engineers, and developer advocates. Developer advocates reported increased productivity, and metrics related to external developer satisfaction increased significantly.

- Acted as the team's interim Engineering Manager for four months. Collaborated with Product Manager on team roadmap planning, directed the work of eight engineers, mentored team members through regular one-on-one meetings.
- Conducted hundreds of technical and leadership interviews, regular member on hiring committees.

Staff Software Engineer, Payment API and Alternative Payment Methods 2018-2022

- Lead backend engineer for initial development of Tap to Pay on iPhone, which started out as an experimental skunkworks project and has grown into a core offering that processes billions of dollars of gross payments volume. Coordinated development across server and mobile engineering disciplines, collaborated extensively with partner's engineering team and made key contributions to strategic decisions. Developed two new highly-available service components (99.99% SLO) and integrated them seamlessly into existing systems. Advised teams on subsequent features.
- Authored numerous engineering designs and led implementation for payment methods and features in first party products and public APIs, including Cash App Pay.

McKinsey & Co.

Digital Expert 2016-2018

- Tech lead and staff engineer for "internal start-up" incubators within established companies in multiple industries. Built engineering teams and delivered full-stack MVPs.
- Implemented a dynamic pricing engine for a specialty e-commerce retailer, resulting in millions of dollars of annual margin gains.
- Led development of a configurable, auditable calculation engine for flagship financial analytics application as part of re-architecture and cloud migration of a legacy system.

US Department of Defense

Senior Software Developer 2013-2016

- Developed *Introduction to Python Programming* course used by more than 2000 employees, providing a mix of traditional in-person lessons and self-paced interactive exercises.
- Developed and delivered web applications to increase efficiency and effectiveness of various subject matter experts: intelligence analysts, signals analysts, and big data engineers.
- Collaborated with intelligence analysts to develop map-reduce analytics that using advanced statistical techniques and were deployed to large-scale Hadoop clusters.

Applied Research Mathematician 2010-2013

- Proposed optimizations for advanced cryptographic algorithms and developed prototype implementation to validate the effectiveness of the proposals.
- Introduced novel integer linear programming approaches in large-scale data processing systems.

Education

The University of Texas at Austin, Ph.D. in Mathematics 2010

Dissertation: *An Analogue of the Korkin-Zolotarev Lattice Reduction for Vector Spaces over Number Fields*

Awards: 2004 VIGRE Fellowship, 2010 Texas Exes Teaching Award

Williams College, B.A. in Mathematics and Russian Literature 2003

Honors: *Cum laude*, with highest honors in Mathematics

Extracurricular activities: Varsity rowing team captain, New England Rowing Championships medalist, Head of the Charles Regatta medalist.

Skills

Java, Python, Ruby, Payments, Automated Testing, CI/CD, Agile, TDD, High Availability, Scalability, Reliability, SLOs, SLAs, RAG, LLMs, Generative AI